

AI operating system

Ten diagrams, each proving a different claim about the same system. Every claim was re-audited against the factory's code on 14 September 2026, and every plate was redrawn from the result. Prepared as reference artwork for the public ai-systems repository.

10	245	0	September 2026
plates	claims audited	figures contested	compiled

DERIVED FROM

skill-ecosystem at 5739a97 — *the factory, private, read at a named commit*
four product repositories — *read-only, each at a named commit*
Vercel production deployments — *product status, 14 September 2026*
plans/ais-plate-claims.json — *245 claims, every citation resolved by a script*

Every figure in this set carries its own evidence tier. There is no blanket disclaimer, because a single global caveat tells a reader that something is unverified without telling them which thing, and nothing on any page can then be trusted. The tiers cost more to maintain and are worth it.

Visual grammar

The plates form one system. Colour carries meaning and each colour means exactly one thing; form carries meaning independently of colour. Where a distinction matters, it is drawn as topology rather than hue, so it survives a monochrome print.

COLOUR

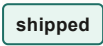
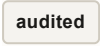
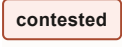
 Teal	Platform substrate: layers, stages, checks, lanes
 Purple	Products, and nothing else
 Coral	Enforcement: gates, halts, drift-prone surfaces
 Gray	Structural: frames, origins, inputs, outputs

FORM

Dashed outline	A gate or boundary condition. Never decorative.
Unit ticks	A count of agents, stages, rounds or attempts. Ticks reconcile by construction.
Numeral	A threshold. Drawn as a number so it cannot be mistaken for a reconciling count.
Direction	On plate 05, outcome. Up and left returns, down terminates, straight continues.
Taper	Distillation. Narrower layers sit above wider ones.
Broken rhythm	A deliberate exception. Plate 08's human gate is the case.

Evidence tiers

Each plate states what its claim rests on. Nothing is tiered higher than its weakest input: a plate whose mechanism is shipped but whose count is audited is labelled for both.

	A named file implements it. Ask to see the file.
	Counted or observed at a named commit, or in a named source on a named date. A count is audited, because no file implements a number.
	Two sources give different values and the conflict is open. Nothing drawn in this set is contested.
	No traceable source. Nothing unverified is drawn.

The ten plates

Each plate proves one claim. Where two would prove the same thing, one of them is wrong — so the set is not ten views of the architecture, it is ten separate arguments about it.

01	The operating system	A system with a build side and a run side, not a folder of scripts.
02	The knowledge substrate	Importance in the corpus is measured, and the measurement is checked.
03	The compiler spine	Orchestration with isolation, not a prompt chain.
04	The contract spine	One source of truth, and the reach of changing it is computed.
05	The verification stack	Four outcomes, and a failure retrying can fix is told apart from one it cannot.
06	Context residency	Budgeting is a measurement, and so is uptake.
07	The hardening loop	Promotion requires a proof of convergence.
08	The learning loops	Two loops, and in both a person decides what becomes permanent.
09	The runtime engine	The packages a product runs on, and the product that runs on them.
10	Two-lane execution	The constraint is scoped, not total.

The re-audit

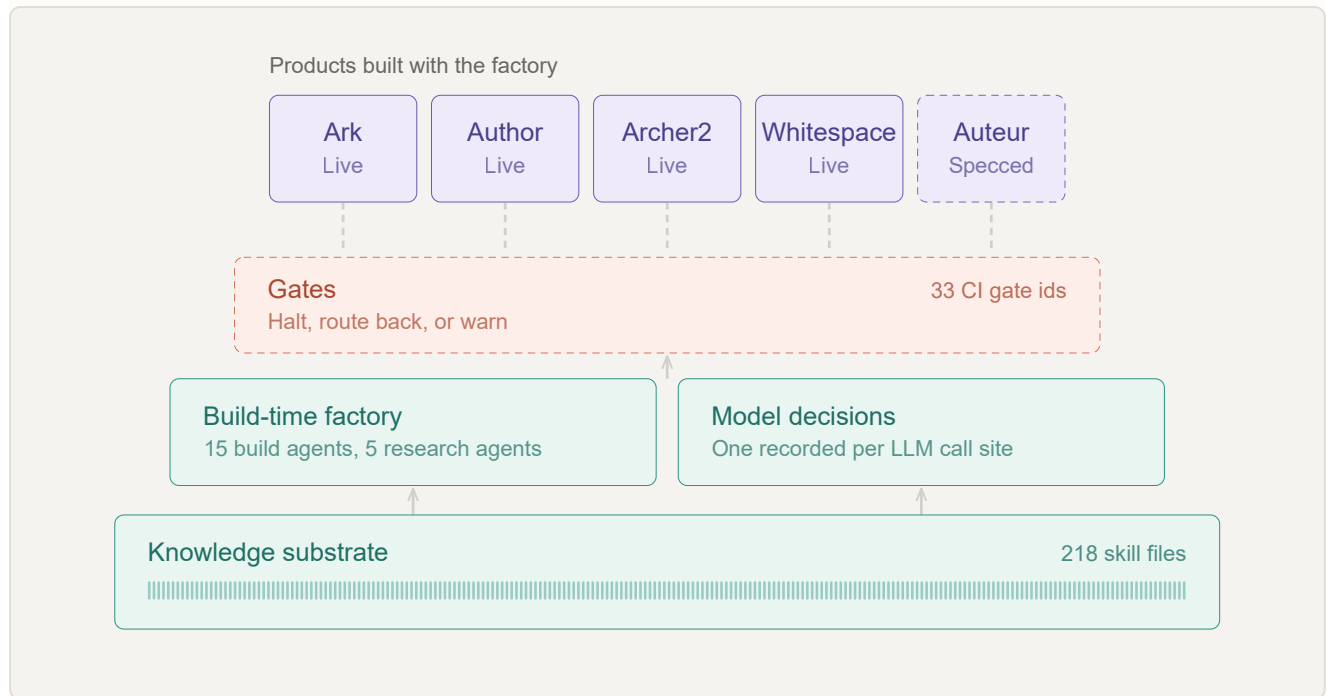
The previous set was drawn from notes and a repository read. On 14 September every claim it made was checked against the factory's code: of 245 claims, 98 held, 127 were corrected, 9 were removed and 11 were added. Most rows below are counts that were contested and are now measured. The rest are mechanisms the plates drew that the code does not have.

FIGURE	WAS	NOW	FROM
Skill files	152, contested	218	code
CI gates	30 against 19	33 gate ids; 19 preflight features	code
Level-0 hubs	11	13 measured, 14 in the set	code
Architecture layers	8 or 9	9	code
Dispatch	a linear chain	12 waves, at most 2 at once	code
Contracts and schemas	19 against 43	48 and 48, one to one	code
Verification order	as described	as the code runs it	code
Context reduction	about 70%	196K-521K to 7K-31K	commit
Hardening exits	clean rounds or a cap	declare hot: yes, extend or stop	code
Unapproved learning tiers	described as applying	not in force	ruling
Cost governance	circuit breakers	none; deltas that only flag	code
MCP connectors	six	three in the default lineup	code
Product on the gateway	Whitespace Hunter	Author	code
Archer2	dormant	live	Vercel

The audit also found defects in the factory itself: a producer agent was handed no prompt the drift audit could read, a named skill section loaded the whole file, three connector tools called actor paths that do not exist, and agent 11's declare-hot question was specified and never asked. Each was fixed, with a check that fails without the fix, before the plates were redrawn from the fixed code. Where the code was wrong, the drawing did not move to hide it.

The operating system

A system with a build side and a run side, not a folder of scripts.



READING

The taper is the argument: one corpus of skill files carries the factory that builds products and the call-site discipline for its model calls. Narrower layers sit above wider ones, so the compression from the corpus to the products is visible rather than asserted.

CHANGED IN THIS REVISION

The counts are re-measured and no longer contested: 218 skill files and 33 CI gate ids. Archer2 is live, and the product band reads four built and one specced.

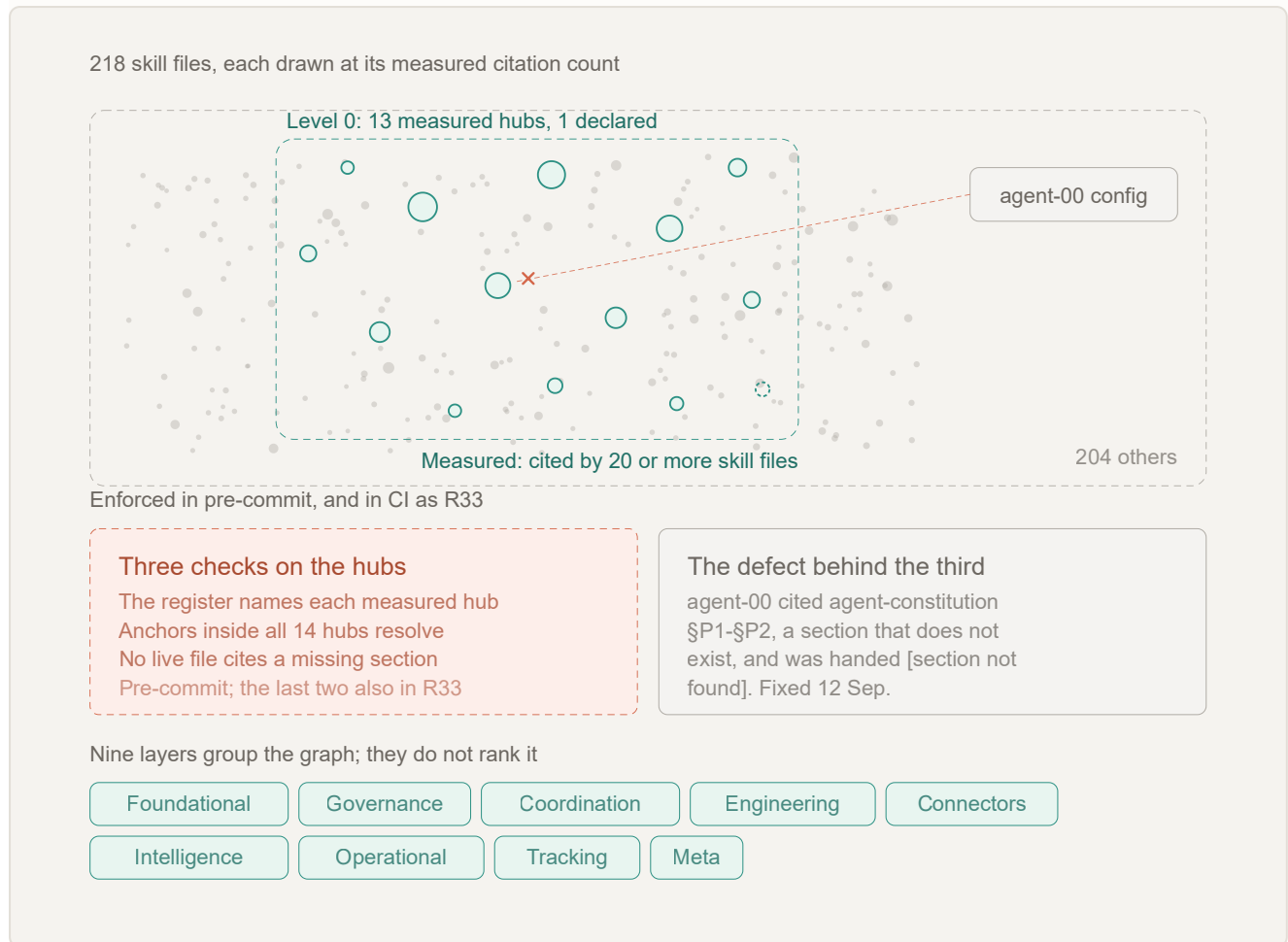
EVIDENCE

audited 218 files, 20 agents, 33 CI gate ids

audited product status, 14 Sep

The knowledge substrate

Importance in the corpus is measured, and the measurement is checked.



READING

A skill file is a measured hub when twenty or more other skill files cite it, and every node is drawn at the size of its own count. Three files are declared by hand, so the Level-0 set is fourteen, and pre-commit checks hold the hubs' names and anchors.

CHANGED IN THIS REVISION

Every node is now sized by its measured count, and the edges are gone, because the factory records how often a skill is cited, not by whom. The defect edge starts at agent 00's config, and the panel names the checks that catch its class.

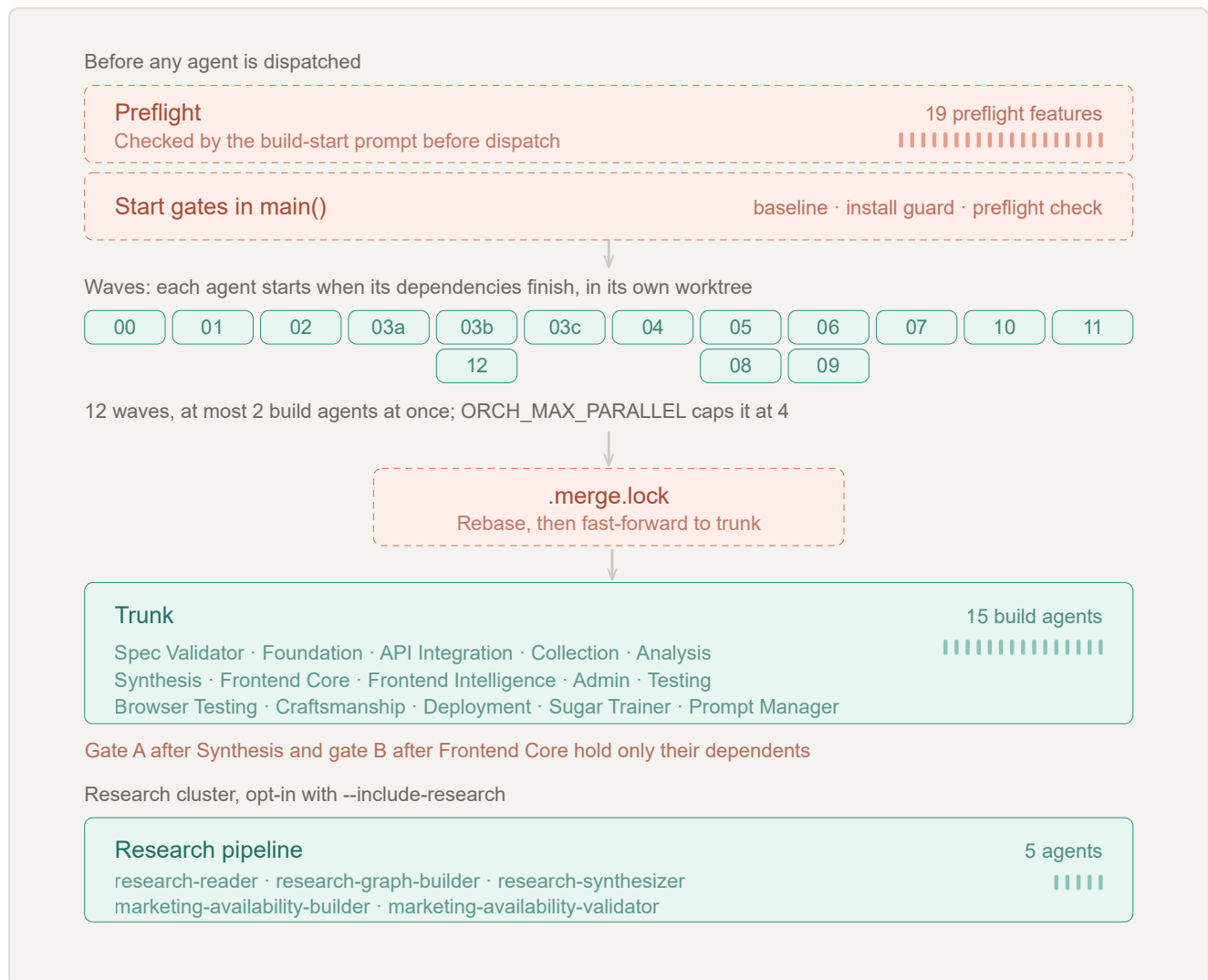
EVIDENCE

shipped skill-ref-count.ts, verify-hub-integrity.mjs, audit-hub-citations.mjs

audited 13 measured hubs, a set of 14

The compiler spine

Orchestration with isolation, not a prompt chain.



READING

Every agent whose dependencies have finished starts at once, up to four, each in its own git worktree, and results reach trunk one at a time under a file lock. The strip is the build graph's real twelve waves.

CHANGED IN THIS REVISION

Redrawn from a chain to waves: the scheduler that runs agents together landed after the plate was drawn. The orchestrator's own start gates now sit beside the nineteen preflight features.

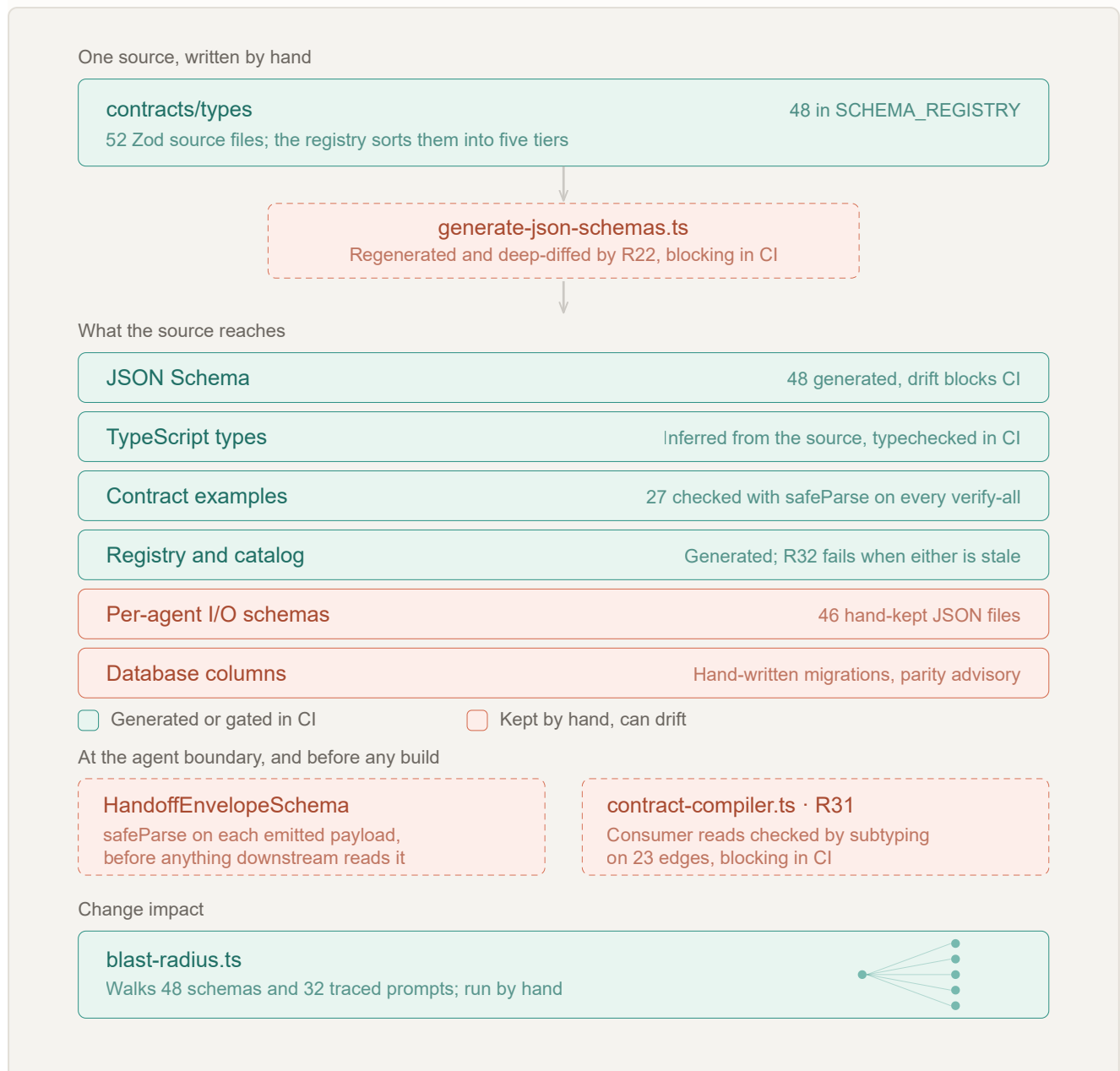
EVIDENCE

shipped agent-scheduler.ts, merge-lock.ts, review-gate.ts

audited 12 waves, at most 2 at once

The contract spine

One source of truth, and the reach of changing it is computed.



READING

Zod is the source. What is generated is gated in CI, and what is kept by hand is drawn in coral because it can drift. Each emitted handoff is parsed against its boundary schema, and the contract compiler checks declared reads by structural subtyping.

CHANGED IN THIS REVISION

Forty-eight contracts generate forty-eight schemas, one to one, so there is no ratio to draw. contract-compiler.ts is now a blocking CI gate, and the blast-radius frontier is the widest the registry holds.

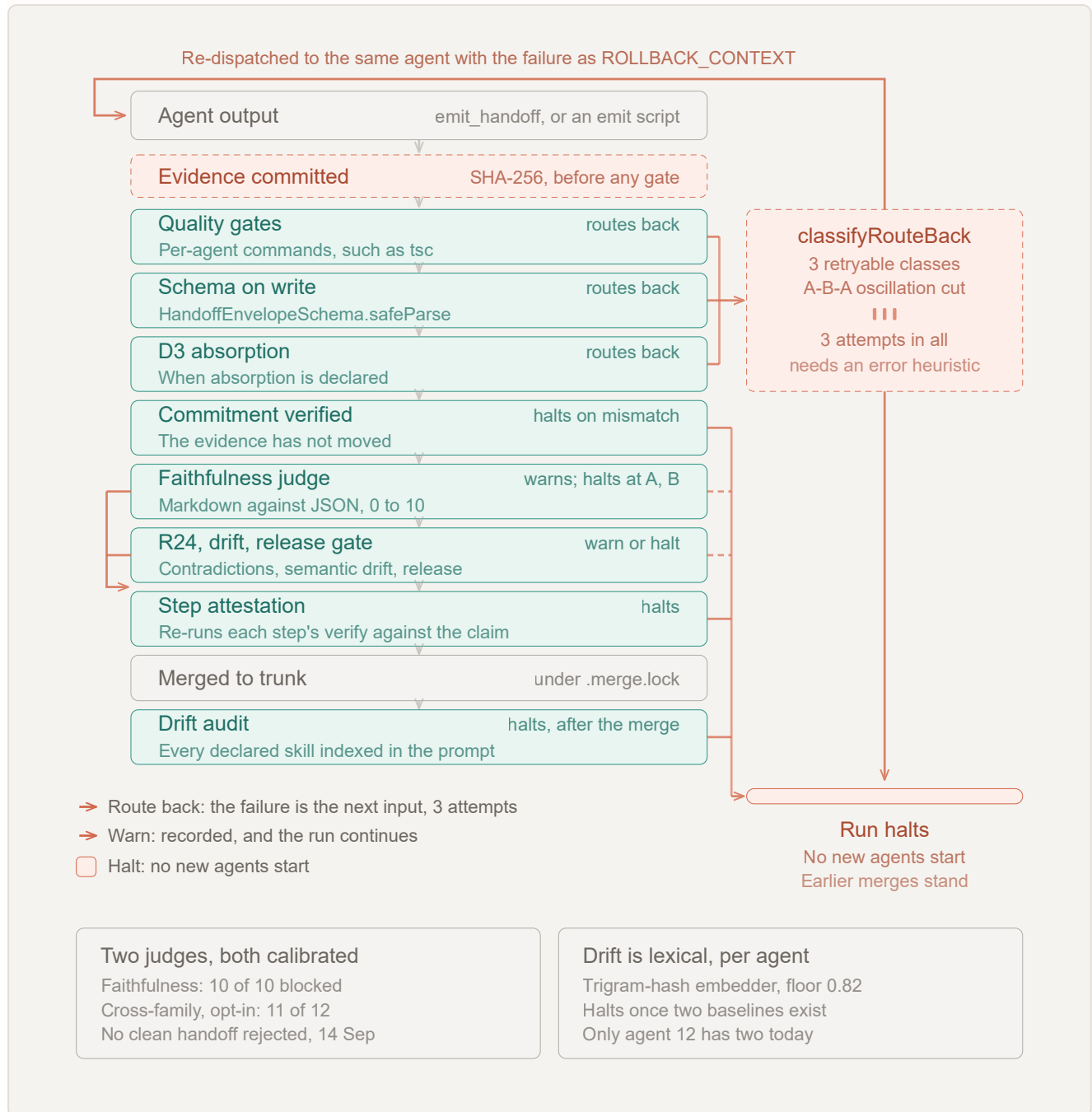
EVIDENCE

shipped R22, R31, R32, HandoffEnvelopeSchema

audited 48 contracts, 46 hand-kept schemas

The verification stack

Four outcomes, and a failure retrying can fix is told apart from one it cannot.



READING

Direction encodes outcome: up and to the left returns, down terminates, straight continues. Evidence is committed when an agent emits. Only the first three checks can route back, and every later failure halts.

CHANGED IN THIS REVISION

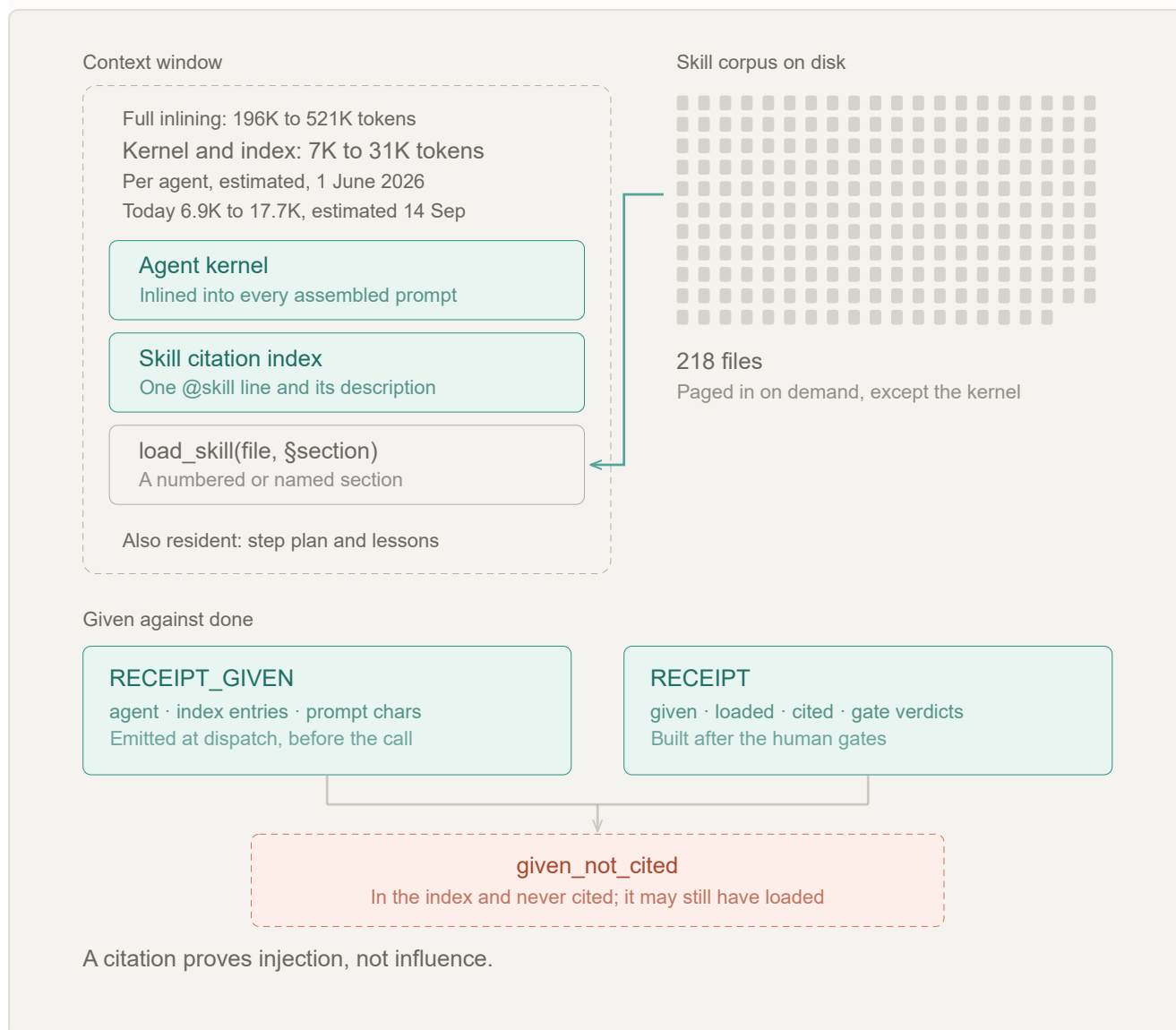
The most corrected plate in the set. It now draws the checks in the order the code runs them, the evidence commitment at the entrance, and the drift audit after the merge; a retry also needs a matching error heuristic.

EVIDENCE

shipped route-back.ts, evidence-commitment.ts, semantic-drift.ts, attestation-verifier.ts

Context residency

Budgeting is a measurement, and so is uptake.



READING

What sits inside the window and what waits outside it. Token reduction is a cost claim; receipts make the harder one, recording what was given against what the agent cited, and the receipt contract says a citation proves injection, not influence.

CHANGED IN THIS REVISION

The reduction is the recorded one, 196K to 521K down to 7K to 31K tokens per agent, beside today's loads. A named section now loads only that section.

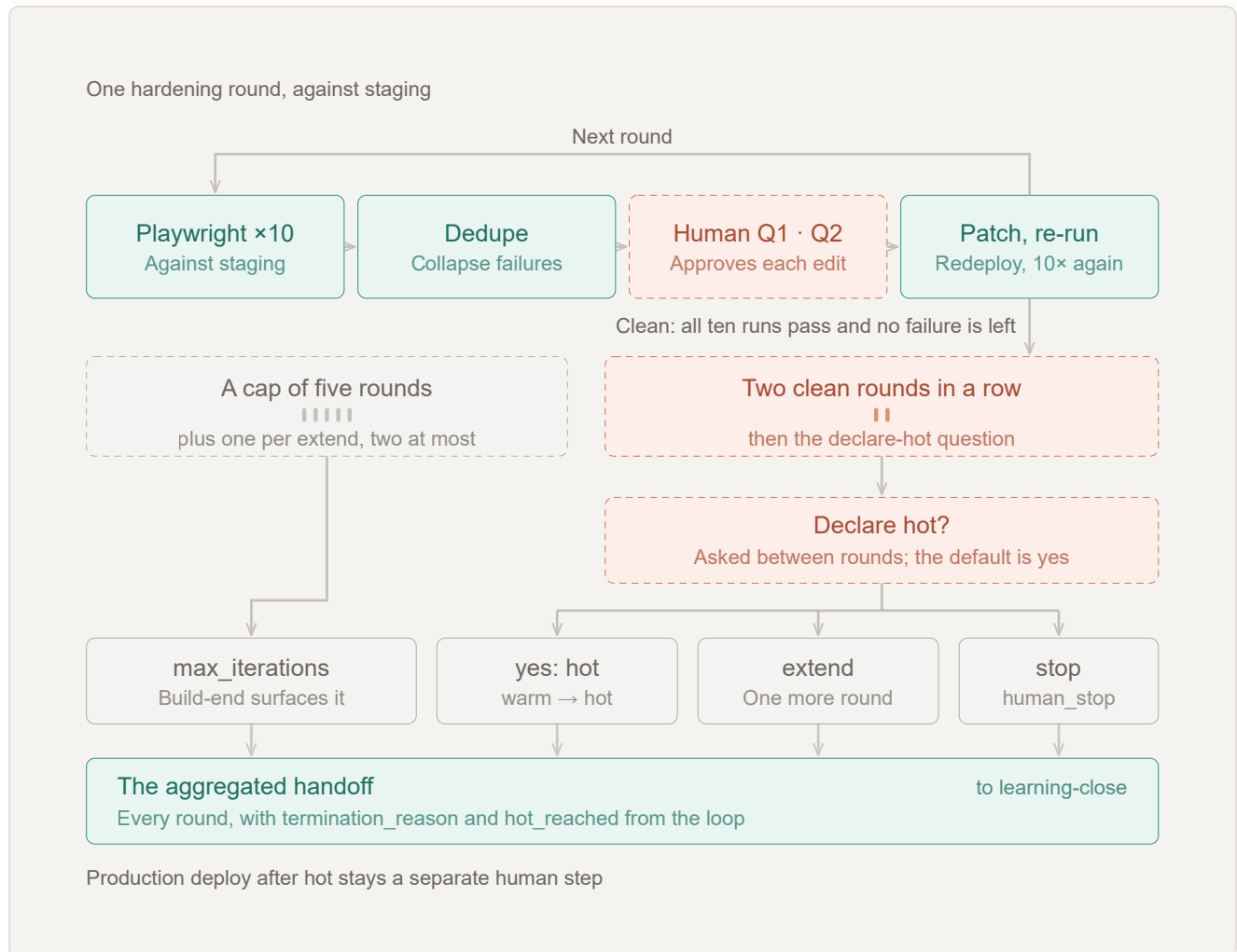
EVIDENCE

shipped load_skill, skill-sections.ts, receipt.ts

audited the reduction and today's loads, estimated

The hardening loop

Promotion requires a proof of convergence.



READING

Two clean rounds in a row lead to the declare-hot question, which the orchestrator asks between rounds. Yes ends hot, stop ends the loop, extend runs one more round, and the cap moves with each extension.

CHANGED IN THIS REVISION

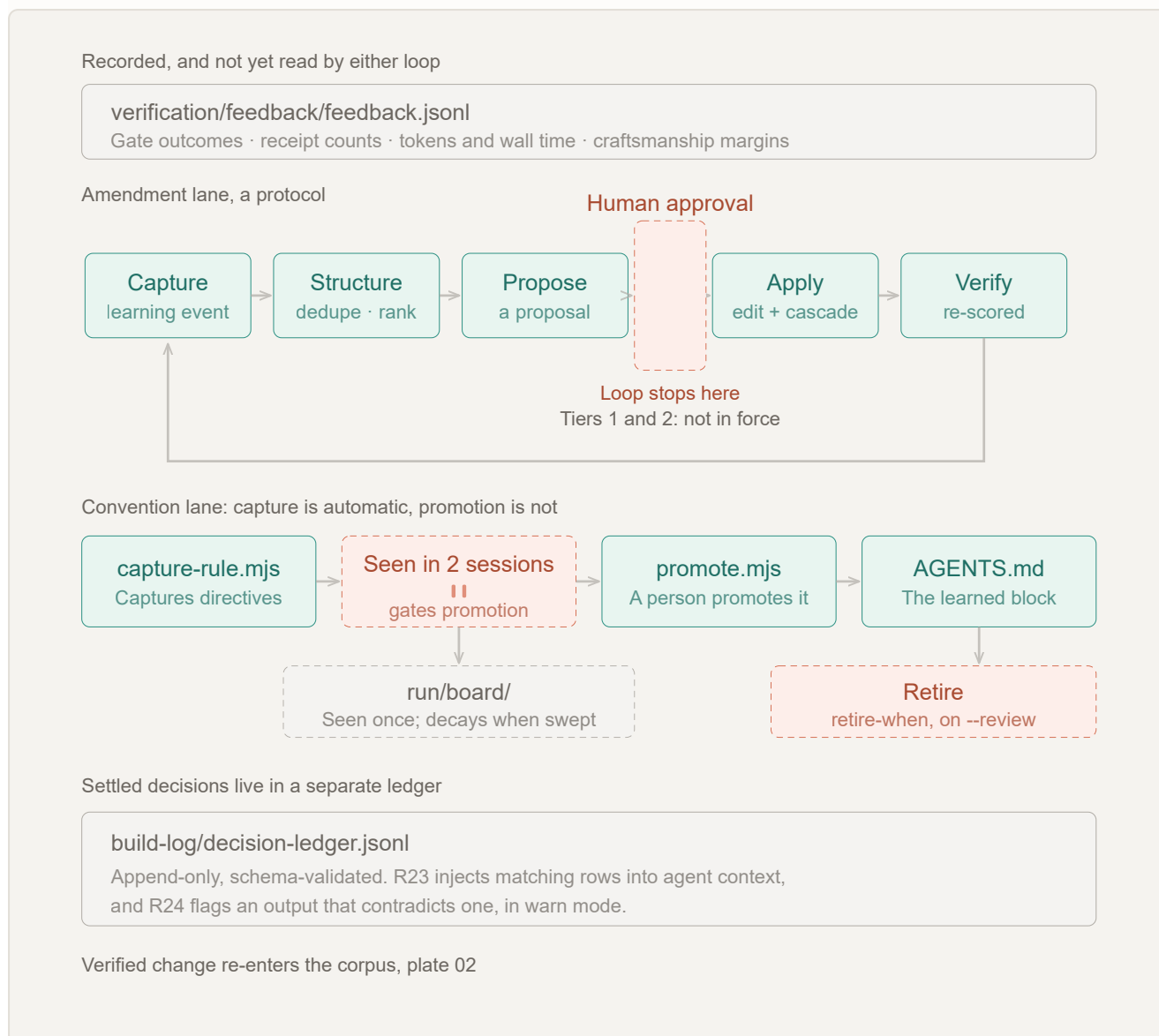
The declare-hot question and the stop condition were specified and never built. They are built now, and the plate draws all four endings and the handoff that carries them.

EVIDENCE

`shipped` `hardening-state.ts`, `askDeclareHot`, `agent-11-steps.md`

The learning loops

Two loops, and in both a person decides what becomes permanent.



READING

A rule that recurs across two sessions is promoted by a person, and a finding seen once decays on the board. The amendment lane's human approval is the only route, and the decision ledger is a third store the pipeline reads back.

CHANGED IN THIS REVISION

The build telemetry is drawn as stored rather than read, because no loop reads it yet. The tiers that would skip the human gate are marked not in force, by operator ruling.

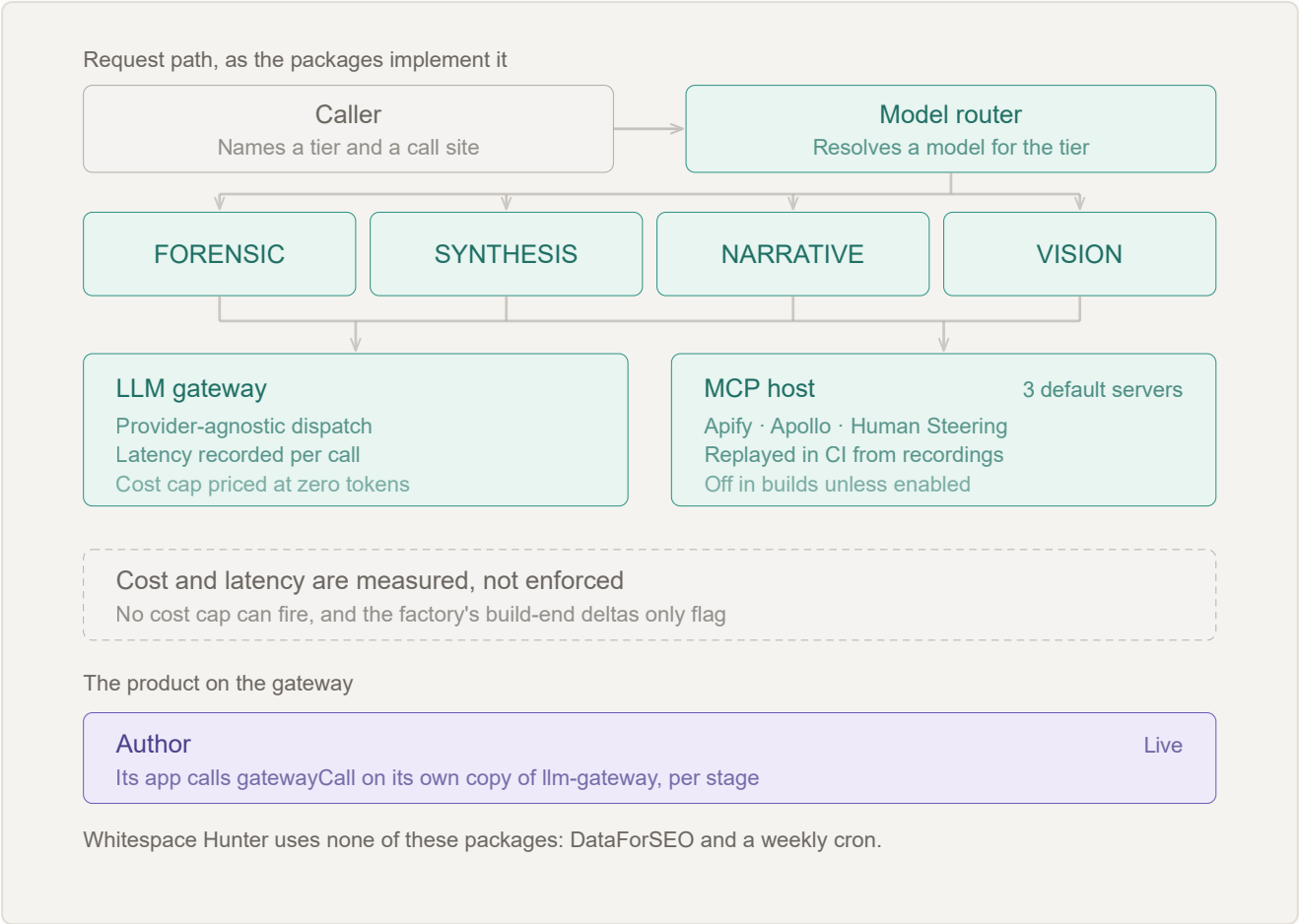
EVIDENCE

shipped capture-rule.mjs, promote.mjs, AGENTS.md, decision-ledger.jsonl

audited the amendment lane, a protocol

The runtime engine

The packages a product runs on, and the product that runs on them.



READING

The caller names a tier, the router resolves a model and the gateway dispatches it. Cost does not stop a run. Author's app is the production caller; Whitespace Hunter, the product with a public URL, uses none of this.

CHANGED IN THIS REVISION

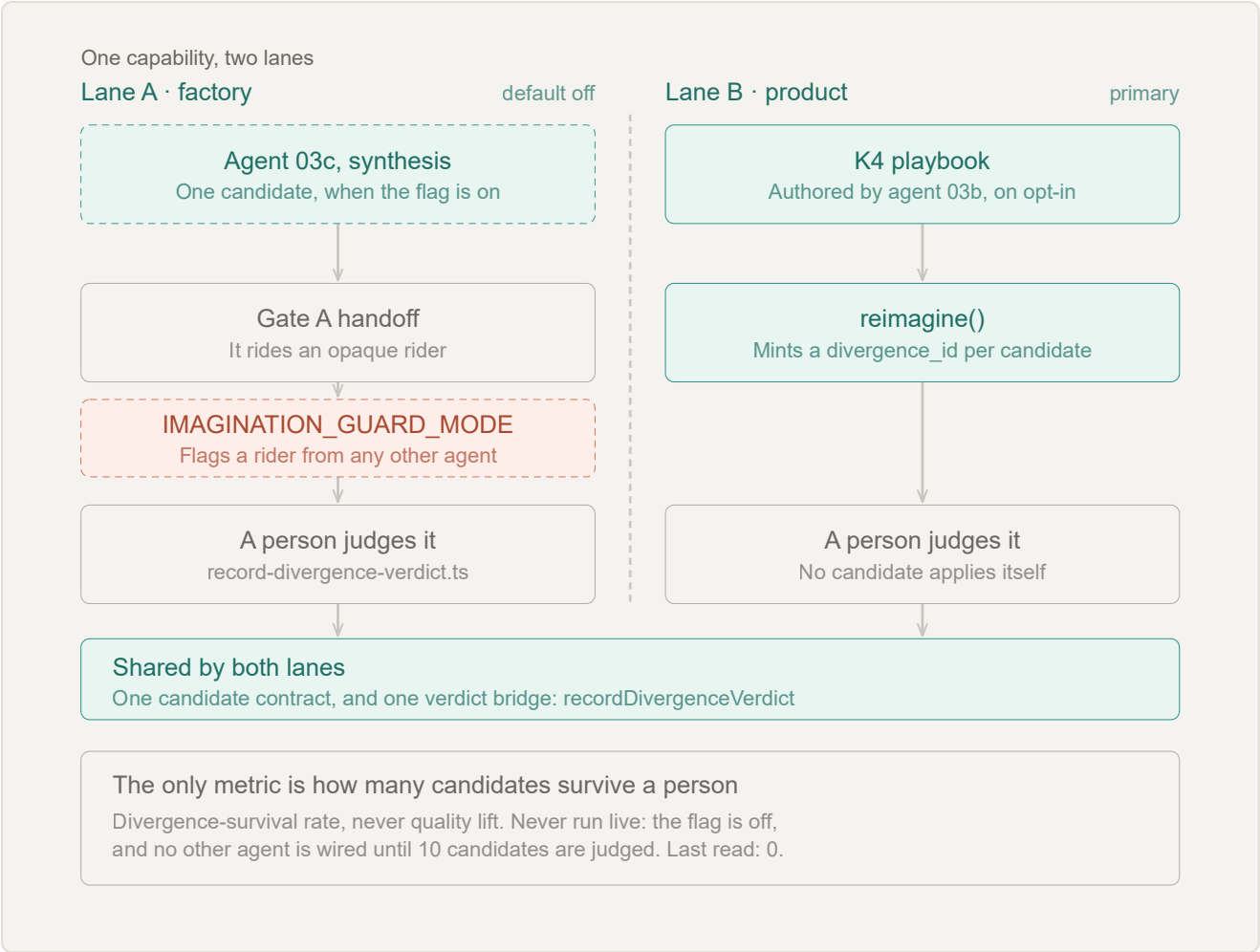
Rebuilt from what exists. Session context, a load estimator, complexity routing and cost circuit breakers were drawn and do not exist, and the product on the gateway moves from Whitespace Hunter to Author.

EVIDENCE

- shipped llm-gateway, model-router, mcp-infra
- audited Author's gateway client

Two-lane execution

The constraint is scoped, not total.



READING

Divergence is allowed in two fenced places: one flag-gated candidate from agent 03c at gate A, and an opt-in product runtime. Both share one candidate contract and one verdict path, and a person judges every candidate.

CHANGED IN THIS REVISION

The implementing files are named, and the tier is shipped. What travels is the whole candidate as an opaque rider, not an identifier under a parity check, and the two lanes share a contract.

EVIDENCE

shipped divergence-candidate.ts, applyImaginationFlag, validate-handoff.ts

audited never run live

Structure

The counts that carry the structural argument, each measured at factory commit 5739a97. A count is audited, not shipped, because no file implements a number; the audit records how each one was measured.

FIGURE	VALUE	TIER	SOURCE	CHANGED
Skill files	218	audited	skills/, per-layer indexes excluded	was 152, contested
In a manifest, parked	170, 48	audited	agent configs, retired-skills.json	new
Architecture layers	9	audited	architecture_layer in frontmatter	was 8 or 9
Agents	20: 15 build, 5 research	audited	configs/agent-*.config.ts	unchanged
Build waves	12, at most 2 at once	audited	dependsOn, the scheduler's rule	new
CI gate ids	33	audited	verify-all.ts	was 30, contested
Preflight features	19	audited	prompts/feature-descriptor.md	were called gates
Registered contracts	48, in five tiers	audited	SCHEMA_REGISTRY	was 19, contested
Generated JSON Schemas	48	audited	one per registry entry	was 43, contested
Per-agent I/O schemas	46	audited	contracts/schemas/agents	new
Measured Level-0 hubs	13, in a set of 14	audited	skill-ref-counts.json	was 11
Static prompt load	6,926 to 17,684 tokens	audited	token-budget-baseline.json	new

Mechanism

Every mechanism the set draws and the file that implements it. The amendment lane is the one row that is a protocol rather than code.

MECHANISM	WHAT IT DOES	IMPLEMENTED IN	TIER
Dependency-wave dispatch	Starts each agent once its dependencies finish, up to 4	agent-scheduler.ts	shipped

MECHANISM	WHAT IT DOES	IMPLEMENTED IN	TIER
Merge serialisation	Rebase, then fast-forward, under .merge.lock	merge-lock.ts	shipped
Schema on write	safeParse on each emitted payload	orchestrator	shipped
Contract compiler	Declared reads checked by structural subtyping (R31)	contract-compiler.ts	shipped
Change impact	Breadth-first over the registry's imports; run by hand	blast-radius.ts	shipped
Evidence commitment	SHA-256 when an agent emits, verified before any judge	evidence-commitment.ts	shipped
Route-back	3 classes, a matching heuristic, A-B-A cut, 3 attempts	route-back.ts (classifyRouteBack)	shipped
Faithfulness judge	Narrative against JSON; warns, blocks at gates A and B	verify-handoff-faithfulness.ts	shipped
Semantic drift	Trigram-hash embedder, floor 0.82, enforcing at two baselines	semantic-drift.ts	shipped
Step attestation	Re-runs each step's verify; blocking	attestation-verifier.ts	shipped
Drift audit	Every declared skill indexed in the prompt; after the merge	drift-audit.ts	shipped
Section loading	A numbered or named section, never a silent whole file	skill-sections.ts	shipped
Receipts	RECEIPT_GIVEN at dispatch, RECEIPT after the gates	receipt.ts	shipped
Hardening termination	Two clean rounds, then declare hot: yes, extend or stop	hardening-state.ts, askDeclareHot	shipped
Convention promotion	Two sessions, or a recorded reason, by a person	promote.mjs, AGENTS.md	shipped
Decision ledger	Append-only, schema-validated, injected by R23	decision-ledger.jsonl	shipped
Amendment lane	A protocol whose human approval is the only tier in force	learning-loop.md	audited
LLM gateway	Provider adapters and per-call latency; its cost cap cannot fire	llm-gateway	shipped
MCP host	Three default servers, replayed in CI	mcp-infra	shipped
Imagination lanes	One flag-gated candidate at gate A; an opt-in product runtime	runtime-imagination, validate-handoff.ts	shipped

Products

Built with the factory. Status is audited from each project's Vercel production deployments on 14 September 2026; the repositories are private.

PRODUCT	WHAT IT IS	STATUS
Ark	A brand book researched and written from a URL	Live; deployed 7 Sep
Author	A content playbook compiler, from a brand's URL	Live; deployed 26 Jul
Archer2	B2B sales intelligence for a target account	Live; deployed 26 Jul
Whitespace Hunter	Runway scoring for emerging categories, at whitespace-hunter.vercel.app	Live since 1 Sep
Auteur	AI creative production	Specced

Not drawn

FIGURE	PREVIOUS VALUE	WHY IT IS OUT
Residency class counts	12 resident / 60 index / 132 paged	No source, and the counts summed to the wrong total.
Build-time compression	2-4 weeks against 6-9 months	A comparative claim with no measured baseline on either side.
Intervention reduction	up to 80% fewer manual interventions	No instrument produces this number. The phrase 'up to' is doing all the work.
Trend surge magnitudes	18x, +1800%	One run's output from a real tool, not a headline figure.
Type-check saving	about 6 seconds per dispatch	Not measured, and no plate makes a claim it would support.
Graded prompt corpus	471	Not re-counted in this audit.

One defect is recorded here because it is worth publishing rather than despite being embarrassing. `agent-00.config.ts` cited a section anchor that did not exist, so the spec validator — first in the pipeline, the gate every build passes through — was handed the literal string `[section not found]` where its constitution should have been. A citation that silently resolves to an error string is worse than one that throws, because the pipeline keeps running and everything downstream is confidently built on nothing. It was fixed on 12

September, and since 14 September a citation to a section that does not exist fails in pre-commit and in CI.

The plates are compiled, not drawn. A grammar module holds the colour roles, ticks, gates and taper as primitives, and each plate is written in terms of those, so a grammar violation is impossible to express rather than something a reviewer has to catch. It is the argument plate 04 makes about schemas, applied to the artwork. The source and its regeneration instructions are in `tools/diagram-compiler`. The plates are committed as SVG, not raster: they render natively on GitHub, appear in diffs, and let a reviewer watch the architecture change over time.